

PID Control of dual axis tilt rotor balancing setup

Aim of the Experiment

The aim of the experiment is to setup 2 PID controllers that control pitch and yaw. The input is to be given as a set of step inputs i.e. (pitch = 0deg, yaw = 0deg) then (pitch = 15deg, yaw = -25deg), and the design parameters are as follows:-

- Pitch:-
 - Peak overshoot: $\leq 5\%$
 - Peak time: $\leq 3.5s$
- Yaw:-
 - Peak overshoot: $\leq 5\%$
 - Peak time: $\leq 2.5s$

The setpoints should be achieved simulataneously, not separately.

Materials Required

- Arduino Mega 2560
- type-A to type-B USB cable
- PID tuning PCB
- Female-to-Female JST cable
- Dual axis tilt rotor balancing setup
- L298N motor driver module

Block Diagram

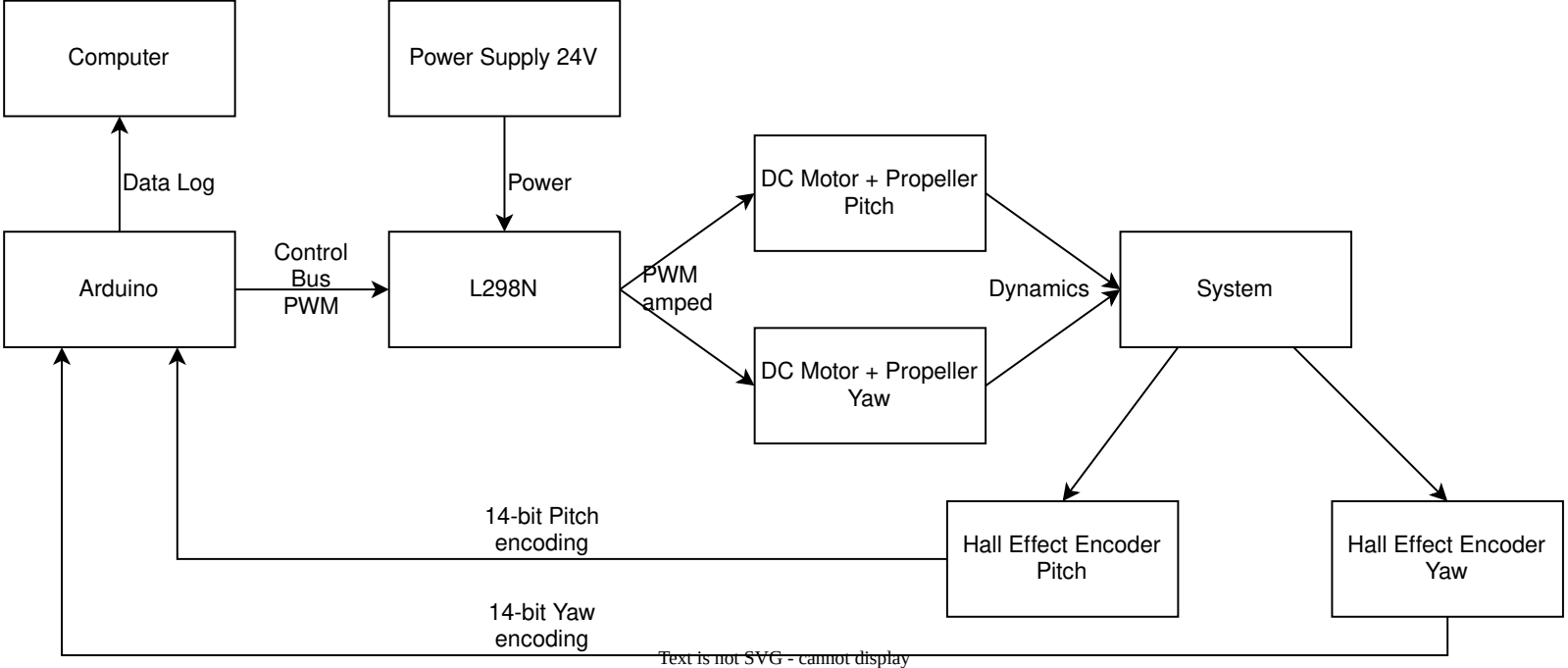


Figure 1: Block Diagram